



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Experiment No. 3

Title: Implement Control Structures and Form Validation; Create a Grading System Based on User-Entered Marks

Software / Tools Required:

1. Visual Studio Code
2. Google Chrome
3. HTML5
4. JavaScript (ES6)

Theory: JavaScript provides control structures that help in decision-making and repetition of tasks. Form validation ensures that user inputs are correct before submitting the data.

a) Control Structures

1. Used to control the flow of program execution.
2. Includes if, if-else, else-if, switch, for, while, and do-while.

b) Form Validation

1. Checks whether user input is valid.
2. Prevents incorrect or incomplete data from being submitted.

c) Conditional Statements

1. Used to compare values and execute specific blocks of code.
2. Helpful in implementing grading systems and validations.

Case Study Implemented:

A password validation webpage was created using JavaScript to verify whether the entered password is valid or not. A grading system was also designed where the user enters marks and the corresponding grade is displayed automatically.

Conclusion:

The practical was completed successfully. Control structures and form validation were implemented using JavaScript. A password validation system and a grading system were developed to validate user input and display the appropriate grade based on the entered marks.